STRONG FORMAL EFFECTIVENESS FOR POLARIZED PROPER FLAT FINITELY PRESENTED SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file quot.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Lemma more-morphisms-lemma-deform-projective (canonical tag 0D4F). cited at proof line 4544. Supplies d_0, finite presentation of A_n, projective homogeneous parts, and system compatibility.
- Lemma more-algebra-lemma-lim-finite-projective-gives-finite-projective (canonical tag 0D4B). cited at proof lines 4565-4566. B_d finite projective and B x_R R_n = A_n.
- Lemma divisors-lemma-relative-proj-flat (canonical tag 0D4C). cited at proof lines 4575-4576. Proj(B) flat over S, L flat.
- Lemma constructions-lemma-base-change-map-proj (canonical tag 01N2). cited at proof lines 4577-4578. Proj commutes with base change, giving the descent isomorphisms.
- Lemma divisors-lemma-relative-proj-proper (canonical tag 0800). cited at proof lines 4588-4589. Properness of X -> S.
- Lemma divisors-lemma-relative-proj-finite-presentation (canonical tag 0D4D). cited at proof lines 4589-4590. Finite presentation of X -> S and of L.
- Lemma more-morphisms-lemma-finite-free-open (canonical tag 0CZR). cited at proof lines 4594-4595. Invertibility locus is an open neighbourhood of X_1.
- Lemma more-algebra-lemma-limit-henselian (canonical tag 0CT7). cited at proof lines 4598-4599 and 4619-4620. Ker(R -> R_1) in the Jacobson radical.
- Lemma more-morphisms-lemma-ample-in-neighbourhood (canonical tag 0D2S). cited at proof lines 4602-4603. Ampleness on fibres is open and contains S_1.
- Lemma algebra-lemma-NAK (canonical tag 00DV). cited at proof line 4618, used again at 4623-4624. Nakayama: Psi surjective and Ker(Psi) generated by the lifted relations.

Source reference. Original lemma 0D4F: sources/stacks-chow/tags.txt, source lines 16450–16450. Cited at proof line 4544: gives d_0 and the finite-presentation properties of each A_n plus compatibility of the system; load-bearing.

Date: stacks-project:quot.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0D4B: sources/stacks-chow/tags.txt, source lines 16446–16446. Cited at proof lines 4565-4566: B_d finite projective over R and B x_R R_n = A_n; load-bearing.

Source reference. Original lemma 0D4C: sources/stacks-chow/tags.txt, source lines 16447–16447. Cited at proof lines 4575-4576: X = Proj(B) flat over S and L flat; load-bearing.

Source reference. Original lemma 01N2: sources/stacks-chow/tags.txt, source lines 2143–2143. Cited at proof lines 4577-4578: canonical isomorphisms X x_S S_n = X_n and L|_{X_n} = L_n; load-bearing.

Source reference. Original lemma 0800: sources/stacks-chow/tags.txt, source lines 10165–10165. Cited at proof lines 4588-4589: B finite presentation over R makes X -> S proper; load-bearing.

Source reference. Original lemma 0D4D: sources/stacks-chow/tags.txt, source lines 16448–16448. Cited at proof lines 4589-4590: X -> S of finite presentation and L finitely presented; load-bearing.

Source reference. Original lemma 0CZR: sources/stacks-chow/tags.txt, source lines 16286–16286. Cited at proof lines 4594-4595: L is invertible on an open neighbourhood of X_1; load-bearing.

Source reference. Original lemma 0CT7: sources/stacks-chow/tags.txt, source lines 16057–16057. Cited at proof lines 4598-4599 and 4619-4620: Ker(R -> R_1) inside the Jacobson radical, making every neighbourhood of X_1 equal to X and enabling Nakayama; load-bearing twice.

Source reference. Original lemma 0D2S: sources/stacks-chow/tags.txt, source lines 16392–16392. Cited at proof lines 4602-4603: ampleness locus open and contains S_1, hence equals S; load-bearing.

Source reference. Original lemma 00DV: sources/stacks-chow/tags.txt, source lines 512–512. Cited at proof line 4618 (and used again at 4623-4624): surjectivity of Psi and generation of its kernel by the lifts G_j; load-bearing twice.

Lemma 0.2 (Strong formal effectiveness for polarized schemes)**.** *Let (R_n) be an inverse system of rings with surjective transition maps whose kernels are locally nilpotent. Set $R = \lim R_n$. Set $S_n = \mathrm{Spec}(R_n)$ and $S = \mathrm{Spec}(R)$. Consider a commutative diagram*

$$\begin{array}{ccccccc} X_1 & \xrightarrow[i_1]{} & X_2 & \xrightarrow[i_2]{} & X_3 & \longrightarrow & \ldots \\ \downarrow & & \downarrow & & \downarrow & & \\ S_1 & \longrightarrow & S_2 & \longrightarrow & S_3 & \longrightarrow & \ldots \end{array}$$

of schemes with cartesian squares. Suppose given $(\mathcal{L}_n, \varphi_n)$ where each $\mathcal{L}_n$ is an invertible sheaf on X_n and $\varphi_n : i_n^*\mathcal{L}_{n+1} \to \mathcal{L}_n$ is an isomorphism. If

(1) *$X_n \to S_n$ is proper, flat, of finite presentation, and*
(2) *$\mathcal{L}_1$ is ample on X_1*

then there exists a morphism of schemes $X \to S$ proper, flat, and of finite presentation and an ample invertible $\mathcal{O}_X$-module $\mathcal{L}$ and isomorphisms $X_n \cong X \times_S S_n$ and $\mathcal{L}_n \cong \mathcal{L}|_{X_n}$ compatible with the morphisms i_n and φ_n.

Proof. Choose d_0 for $X_1 \to S_1$ and $\mathcal{L}_1$ as in More on Morphisms, Lemma 0D4F. For any $n \geq 1$ set

$$A_n = R_n \oplus \bigoplus\nolimits_{d \geq d_0} H^0(X_n, \mathcal{L}_n^{\otimes d})$$

By the lemma each A_n is a finitely presented graded R_n-algebra whose homogeneous parts $(A_n)_d$ are finite projective R_n-modules such that $X_n = \mathrm{Proj}(A_n)$ and $\mathcal{L}_n = \mathcal{O}_{\mathrm{Proj}(A_n)}(1)$. The lemma also guarantees that the maps

$$A_1 \leftarrow A_2 \leftarrow A_3 \leftarrow \ldots$$

induce isomorphisms $A_n = A_m \otimes_{R_m} R_n$ for $n \leq m$. We set

$$B = \bigoplus\nolimits_{d \geq 0} B_d \quad \text{with} \quad B_d = \lim\nolimits_n (A_n)_d$$

By More on Algebra, Lemma 0D4B we see that B_d is a finite projective R-module and that $B \otimes_R R_n = A_n$. Thus the scheme

$$X = \mathrm{Proj}(B) \quad \text{and} \quad \mathcal{L} = \mathcal{O}_X(1)$$

is flat over S and $\mathcal{L}$ is a quasi-coherent $\mathcal{O}_X$-module flat over S, see Divisors, Lemma 0D4C. Because formation of Proj commutes with base change (Constructions, Lemma 01N2) we obtain canonical isomorphisms

$$X \times_S S_n = X_n \quad \text{and} \quad \mathcal{L}|_{X_n} \cong \mathcal{L}_n$$

compatible with the transition maps of the system. Thus we may think of $X_1 \subset X$ as a closed subscheme. Below we will show that B is of finite presentation over R. By Divisors, Lemmas 0800 and 0D4D this implies that $X \to S$ is of finite presentation and proper and that $\mathcal{L} = \mathcal{O}_X(1)$ is of finite presentation as an $\mathcal{O}_X$-module. Since the restriction of $\mathcal{L}$ to the base change $X_1 \to S_1$ is invertible, we see from More on Morphisms, Lemma 0CZR that $\mathcal{L}$ is invertible on an open neighbourhood of X_1 in X. Since $X \to S$ is closed and since $\mathrm{Ker}(R \to R_1)$ is contained in the Jacobson radical (More on Algebra, Lemma 0CT7) we see that any open neighbourhood of X_1 in X is equal to X. Thus $\mathcal{L}$ is invertible. Finally, the set of points in S where $\mathcal{L}$ is ample on the fibre is open in S (More on Morphisms, Lemma 0D2S) and contains S_1 hence equals S. Thus $X \to S$ and $\mathcal{L}$ have all the properties required of them in the statement of the lemma.

We prove the claim above. Choose a presentation $A_1 = R_1[X_1, \ldots, X_s]/(F_1, \ldots, F_t)$ where X_i are variables having degrees d_i and F_j are homogeneous polynomials in X_i of degree e_j. Then we can choose a map

$$\Psi : R[X_1, \ldots, X_s] \longrightarrow B$$

lifting the map $R_1[X_1, \ldots, X_s] \to A_1$. Since each B_d is finite projective over R we conclude from Nakayama's lemma (Algebra, Lemma 00DV using again that $\mathrm{Ker}(R \to R_1)$ is contained in the Jacobson radical of R) that Ψ is surjective. Since $- \otimes_R R_1$ is right exact we can find $G_1, \ldots, G_t \in \mathrm{Ker}(\Psi)$ mapping to $F_1, \ldots, F_t$ in $R_1[X_1, \ldots, X_s]$. Observe that $\mathrm{Ker}(\Psi)_d$ is a finite projective R-module for all $d \geq 0$ as the kernel of the surjection $R[X_1, \ldots, X_s]_d \to B_d$ of finite projective R-modules. We conclude from Nakayama's lemma once more that $\mathrm{Ker}(\Psi)$ is generated by $G_1, \ldots, G_t$. $\square$